

```

import pandas as pd
import matplotlib.pyplot as plt
from sklearn.linear_model import LinearRegression
url = r"C:\Users\CAD LAB 16\Desktop\Data Set\Energy Demand
Forecasting\DOM_hourly.csv"
try:
    data = pd.read_csv(url)
    temp_col = 'DOM_MW'
except Exception as e:
    print("Error loading file:", e)
data['TimeIndex'] = range(len(data))
X = data[['TimeIndex']]
y = data[temp_col]
model = LinearRegression()
model.fit(X, y)
future_steps = pd.DataFrame({
    'TimeIndex': [len(data), len(data) + 1]
})
pred = model.predict(future_steps)
print("Predicted values for next two steps:")
print(pred)

plt.figure(figsize=(12, 6))

plt.plot(data['TimeIndex'], y,
         label='Actual Data',
         alpha=0.5)
plt.plot(data['TimeIndex'],
         model.predict(X),
         color='orange',
         label='Regression Line')
plt.scatter(future_steps['TimeIndex'],
           pred,
           color='red',
           zorder=5,
           label='Future Prediction')
plt.xlabel("Time Index")
plt.ylabel(temp_col)
plt.title("Energy Demand Forecasting")

plt.legend()
plt.grid(True)

plt.show()

```